

in $\mathcal{P}_\ell(D)$. We introduce the space of piecewise-polynomial functions

$$\mathcal{P}_\ell(\mathcal{T}_h) := \{v \in L^2(\mathcal{T}_h) : v|_K \in \mathcal{P}_\ell(K), \forall K \in \mathcal{T}_h\}$$

with respect to the partition $\mathcal{T}_h$ and the space of piecewise-polynomial functions

$$\mathcal{P}_\ell(\mathcal{F}_h) := \{\hat{v} \in L^2(\mathcal{F}_h) : \hat{v}|_F \in \mathcal{P}_\ell(F), \forall F \in \mathcal{F}_h\}$$

with respect to the skeleton $\mathcal{F}_h$ of the mesh $\mathcal{T}_h$. The subspace of $L^2(\mathcal{T}_h, \mathbb{R}^{m \times n})$ with components in $\mathcal{P}_\ell(\mathcal{T}_h)$ is denoted $\mathcal{P}_\ell(\mathcal{T}_h, \mathbb{R}^{m \times n})$. Likewise, $\mathcal{P}_\ell(\mathcal{F}_h, \mathbb{R}^{m \times n})$ stands for the subspace of $L^2(\mathcal{F}_h, \mathbb{R}^{m \times n})$ with components in $\mathcal{P}_\ell(\mathcal{F}_h)$. We finally consider

$$\mathcal{P}_\ell(\partial\mathcal{T}_h, \mathbb{R}^{m \times n}) := \left\{ \phi \in L^2(\partial\mathcal{T}_h, E); \phi|_{\partial K} \in \mathcal{P}_\ell(\partial K, \mathbb{R}^{m \times n}), \forall K \in \mathcal{T}_h \right\},$$

where $\mathcal{P}_\ell(\partial K, \mathbb{R}^{m \times n}) := \prod_{F \in \mathcal{F}(K)} \mathcal{P}_\ell(F, \mathbb{R}^{m \times n})$.

REMARK 2. *It is important to keep in mind that, by definition, the functions in $L^2(\partial\mathcal{T}_h, \mathbb{R}^{m \times n})$ and $\mathcal{P}_\ell(\partial\mathcal{T}_h, \mathbb{R}^{m \times n})$, are multi-valued on every interior face F , whereas the functions in $L^2(\mathcal{F}_h, \mathbb{R}^{m \times n})$ and $\mathcal{P}_\ell(\mathcal{F}_h, \mathbb{R}^{m \times n})$ are single-valued on each face F .*

We consider $\mathbf{n} \in \mathcal{P}_0(\partial\mathcal{T}_h, \mathbb{R}^d)$, where $\mathbf{n}|_{\partial K} = \mathbf{n}_K$ is the unit normal vector of ∂K oriented toward the exterior of K . Obviously, if $F = K \cap K'$ is an interior edge/face of $\mathcal{F}_h$, then $\mathbf{n}_K = -\mathbf{n}_{K'}$ on F . If $\boldsymbol{\varphi} \in H^s(\mathcal{T}_h, \mathbb{R}^{m \times n})$, with $s > 1/2$, the function $\boldsymbol{\varphi}|_{\partial\mathcal{T}_h} \in L^2(\partial\mathcal{T}_h, \mathbb{R}^{m \times n})$ is meaningful by virtue of the trace theorem. For the same reason, if $\boldsymbol{\varphi} \in H^1(\Omega, \mathbb{R}^{m \times n})$ then $\hat{\boldsymbol{\varphi}} := \boldsymbol{\varphi}|_{\mathcal{F}_h}$ is well defined in $L^2(\mathcal{F}_h, \mathbb{R}^{m \times n})$.

For $k \geq 0$, we introduce the finite-dimensional subspaces of $\mathcal{H}_1$ and $\mathcal{H}_2$ given by

$$\mathcal{H}_{1,h} := \mathcal{P}_{k+1}(\mathcal{T}_h, \mathbb{R}^{d \times 2}) \quad \text{and} \quad \mathcal{H}_{2,h} := \mathcal{P}_k(\mathcal{T}_h, \mathbb{R}_{\text{sym}}^{d \times d}) \times \mathcal{P}_k(\mathcal{T}_h),$$

respectively.

We consider the following discrete trace inequality.

LEMMA 2. *There exists a constant $C > 0$ independent of h and k such that*

$$(31) \quad \left\| \frac{h_{\mathcal{F}}^{1/2}}{k+1} q \right\|_{0, \partial\mathcal{T}_h} \leq C \|q\|_{0, \mathcal{T}_h} \quad \forall q \in \mathcal{P}_k(\mathcal{T}_h).$$

Proof. See [22, Lemma 3.2]. □

For any integer $\ell \geq 0$ and $K \in \mathcal{T}_h$, we denote by Π_K^ℓ the $L^2(K)$ -orthogonal projection onto $\mathcal{P}_\ell(K)$. The global projection $\Pi_{\mathcal{T}}^\ell$ in $L^2(\mathcal{T}_h)$ onto $\mathcal{P}_\ell(\mathcal{T}_h)$ is then given by $(\Pi_{\mathcal{T}}^\ell v)|_K = \Pi_K^\ell(v|_K)$ for all $K \in \mathcal{T}_h$. Similarly, the global projection $\Pi_{\mathcal{F}}^\ell$ in $L^2(\mathcal{F}_h)$ onto $\mathcal{P}_\ell(\mathcal{F}_h)$ is given, separately for all $F \in \mathcal{F}_h$, by $(\Pi_{\mathcal{F}}^\ell \hat{v})|_F = \Pi_F^\ell(\hat{v}|_F)$, where Π_F^ℓ is the $L^2(F)$ orthogonal projection onto $\mathcal{P}_\ell(F)$. In the following, we maintain the notation $\Pi_{\mathcal{T}}^\ell$ to refer to the L^2 -orthogonal projection onto $\mathcal{P}_\ell(\mathcal{T}_h, \mathbb{R}^{m \times n})$. It should be noted that the tensorial version of $\Pi_{\mathcal{T}}^\ell$ inherently preserves the symmetry of the matrices, as it is derived by applying the scalar operator component-wise. Similarly, we will also use $\Pi_{\mathcal{F}}^\ell$ to denote the L^2 orthogonal projection onto $\mathcal{P}_\ell(\mathcal{F}_h, \mathbb{R}^{m \times n})$.

In the remainder of this section, we provide approximation properties for the projectors defined above. A detailed proof of these results can be found in [22, Section 3] and the references therein.

LEMMA 3. *There exists a constant $C > 0$ independent of h and k such that*

$$(32) \quad \|q - \Pi_{\mathcal{T}}^k q\|_{0, \mathcal{T}_h} + \left\| \frac{h_{\mathcal{F}}^{1/2}}{k+1} (q - \Pi_{\mathcal{T}}^k q) \right\|_{0, \partial\mathcal{T}_h} \leq C \frac{h_K^{\min\{r, k\}+1}}{(k+1)^{r+1}} \|q\|_{1+r, \Omega},$$